Deep Learning (Spring 2026)
Solution for Assignment 4

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Solution 1. False.

$\mathrm{KL}(q_\theta \| p)$ is the *exclusive* (reverse) KL divergence:

$$\mathrm{KL}(q_\theta \| p) = \int q_\theta(x) \log \frac{q_\theta(x)}{p(x)}\, dx.$$

Whenever $q_\theta(x) > 0$ but $p(x) \approx 0$, the integrand blows up, so the optimizer avoids placing mass where p has no support. However, wherever $p(x) > 0$ but $q_\theta(x) = 0$, the integrand contributes 0 (weighted by q_θ), so there is *no* penalty for missing modes. Therefore, minimizing $\mathrm{KL}(q_\theta \| p)$ produces a *mode-seeking* distribution: q_θ concentrates on one (typically the dominant) mode of p rather than uniformly covering all modes. The mode-covering behavior would instead correspond to minimizing $\mathrm{KL}(p \| q_\theta)$ (the inclusive / forward KL).

Solution 2. True.

In VAE, the latent variable z is sampled from the encoder distribution $q_\phi(z|x)$. This sampling operation is not differentiable, which blocks gradient flow back to the encoder parameters ϕ. The reparameterization trick rewrites $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$, so that the stochasticity is moved to ϵ which does not depend on ϕ. This makes z a deterministic, differentiable function of ϕ, enabling gradients to pass back to the encoder.

Solution 3. False.

The exact posterior $p(z|x) = \frac{p(x|z)p(z)}{p(x)}$ requires computing the marginal likelihood $p(x) = \int p(x|z)p(z)\, dz$, which is generally intractable for complex models (e.g., when $p(x|z)$ is parameterized by a neural network). This intractability is precisely the reason VAE introduces a variational approximation $q_\phi(z|x)$ to approximate the true posterior.

Solution 4. False.

In β-VAE, the objective is:

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta\, \mathrm{KL}(q_\phi(z|x) \| p(z)).$$

A large β puts more weight on the KL term, which pushes $q_\phi(z|x)$ closer to the prior $p(z) = \mathcal{N}(0, I)$. Since the prior has independent (uncorrelated) dimensions, a large β enforces the latent variables to be more *independent* (disentangled) from each other, not correlated.

Solution 5. We need to show that alternating maximization of $\mathcal{F}(\theta, q) = \mathbb{E}_{z\sim q}[\log p_\theta(x, z)] + H(q)$ is equivalent to the EM algorithm.

E-step: We show $q^{(t)} = \arg\max_q \mathcal{F}(\theta^{(t)}, q)$ gives $q^{(t)}(z) = p_{\theta^{(t)}}(z|x)$.

Rewrite $\mathcal{F}$:

$$\begin{aligned}
\mathcal{F}(\theta, q) &= \mathbb{E}_{z\sim q}[\log p_\theta(x,z)] + H(q) \\
&= \sum_z q(z)\log p_\theta(x,z) - \sum_z q(z)\log q(z) \\
&= \sum_z q(z)\log\frac{p_\theta(x,z)}{q(z)} \\
&= \sum_z q(z)\log\frac{p_\theta(z|x)\,p_\theta(x)}{q(z)} \\
&= \sum_z q(z)\log\frac{p_\theta(z|x)}{q(z)} + \log p_\theta(x) \\
&= -\mathrm{KL}(q(z)\|p_\theta(z|x)) + \log p_\theta(x). \qquad (1)
\end{aligned}$$

Since $\log p_\theta(x)$ does not depend on q, maximizing $\mathcal{F}(\theta^{(t)}, q)$ over q is equivalent to minimizing $\mathrm{KL}(q\|p_{\theta^{(t)}}(z|x))$. The KL divergence is non-negative and equals zero if and only if $q(z) = p_{\theta^{(t)}}(z|x)$. Therefore:

$$q^{(t)}(z) = p_{\theta^{(t)}}(z|x).$$

This is exactly the E-step of EM, which computes the posterior $p_{\theta^{(t)}}(z|x)$.

M-step: We show $\theta^{(t+1)} = \arg\max_\theta \mathcal{F}(\theta, q^{(t)})$ is equivalent to $\arg\max_\theta Q(\theta|\theta^{(t)})$.
With $q^{(t)}(z) = p_{\theta^{(t)}}(z|x)$ fixed:

$$\begin{aligned}
\mathcal{F}(\theta, q^{(t)}) &= \mathbb{E}_{z\sim q^{(t)}}[\log p_\theta(x,z)] + H(q^{(t)}) \\
&= \mathbb{E}_{z\sim p_{\theta^{(t)}}(z|x)}[\log p_\theta(x,z)] + H(q^{(t)}).
\end{aligned}$$

Since $H(q^{(t)})$ does not depend on θ:

$$\arg\max_\theta \mathcal{F}(\theta, q^{(t)}) = \arg\max_\theta \mathbb{E}_{z\sim p_{\theta^{(t)}}(z|x)}[\log p_\theta(x,z)] = \arg\max_\theta Q(\theta|\theta^{(t)}).$$

This is exactly the M-step of EM.
Therefore, the alternating maximization of $\mathcal{F}(\theta, q)$ is equivalent to the EM algorithm. ■

Solution 6.

1. **(Gaussian KL Divergence)**
For two d-dimensional Gaussians $\mathcal{N}_0(\mu_0, \Sigma_0)$ and $\mathcal{N}_1(\mu_1, \Sigma_1)$:

$$\begin{aligned}
\mathrm{KL}(\mathcal{N}_0\|\mathcal{N}_1) &= \mathbb{E}_{x\sim\mathcal{N}_0}\left[\log\frac{p_0(x)}{p_1(x)}\right] \\
&= \mathbb{E}_{x\sim\mathcal{N}_0}[\log p_0(x)] - \mathbb{E}_{x\sim\mathcal{N}_0}[\log p_1(x)].
\end{aligned}$$

First term:

$$\begin{aligned}
\mathbb{E}_{x\sim\mathcal{N}_0}[\log p_0(x)] &= \mathbb{E}\left[-\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_0| - \frac{1}{2}(x-\mu_0)^T\Sigma_0^{-1}(x-\mu_0)\right] \\
&= -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_0| - \frac{1}{2}\mathrm{tr}(\Sigma_0^{-1}\Sigma_0) \\
&= -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_0| - \frac{d}{2}.
\end{aligned}$$

Second term:

$$\mathbb{E}_{x\sim\mathcal{N}_0}[\log p_1(x)] = \mathbb{E}\left[-\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_1| - \frac{1}{2}(x-\mu_1)^T\Sigma_1^{-1}(x-\mu_1)\right].$$

For the quadratic term, let $x = \mu_0 + \Sigma_0^{1/2}\epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$:

$$\begin{aligned}
&\mathbb{E}[(x-\mu_1)^T\Sigma_1^{-1}(x-\mu_1)] \\
&= \mathbb{E}[((x-\mu_0)+(\mu_0-\mu_1))^T\Sigma_1^{-1}((x-\mu_0)+(\mu_0-\mu_1))] \\
&= \mathbb{E}[(x-\mu_0)^T\Sigma_1^{-1}(x-\mu_0)] + (\mu_0-\mu_1)^T\Sigma_1^{-1}(\mu_0-\mu_1) \\
&= \mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1-\mu_0)^T\Sigma_1^{-1}(\mu_1-\mu_0).
\end{aligned}$$

where the cross terms vanish because $\mathbb{E}[x-\mu_0] = 0$.

Therefore:

$$\mathbb{E}_{x\sim\mathcal{N}_0}[\log p_1(x)] = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_1| - \frac{1}{2}\left[\mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1-\mu_0)^T\Sigma_1^{-1}(\mu_1-\mu_0)\right].$$

Subtracting:

$$\begin{aligned}
\mathrm{KL}(\mathcal{N}_0\|\mathcal{N}_1) &= -\frac{1}{2}\log|\Sigma_0| - \frac{d}{2} + \frac{1}{2}\log|\Sigma_1| + \frac{1}{2}\mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + \frac{1}{2}(\mu_1-\mu_0)^T\Sigma_1^{-1}(\mu_1-\mu_0) \\
&= \frac{1}{2}\left(\mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1-\mu_0)^T\Sigma_1^{-1}(\mu_1-\mu_0) - d + \log\frac{|\Sigma_1|}{|\Sigma_0|}\right).
\end{aligned}$$

■

2. **(Convexity)**

We use the log-sum inequality. For any $\lambda \in [0,1]$ and discrete distributions p_1, p_2, q_1, q_2 over $\mathcal{Y} = \{1, \ldots, n\}$:

$$\begin{aligned}
&\mathrm{KL}(\lambda p_1 + (1-\lambda)p_2 \| \lambda q_1 + (1-\lambda)q_2) \\
&= \sum_{y=1}^{n}[\lambda p_1(y) + (1-\lambda)p_2(y)]\log\frac{\lambda p_1(y) + (1-\lambda)p_2(y)}{\lambda q_1(y) + (1-\lambda)q_2(y)}.
\end{aligned}$$

By the log-sum inequality, for non-negative reals a_i, b_i $(i = 1, 2)$ with $a_1 + a_2 > 0$ and $b_1 + b_2 > 0$:

$$(a_1+a_2)\log\frac{a_1+a_2}{b_1+b_2} \le a_1\log\frac{a_1}{b_1} + a_2\log\frac{a_2}{b_2}.$$

For each y, let $a_1 = \lambda p_1(y)$, $a_2 = (1-\lambda)p_2(y)$, $b_1 = \lambda q_1(y)$, $b_2 = (1-\lambda)q_2(y)$. Applying the log-sum inequality:

$$\begin{aligned}
&[\lambda p_1(y) + (1-\lambda)p_2(y)]\log\frac{\lambda p_1(y) + (1-\lambda)p_2(y)}{\lambda q_1(y) + (1-\lambda)q_2(y)} \\
&\le \lambda p_1(y)\log\frac{\lambda p_1(y)}{\lambda q_1(y)} + (1-\lambda)p_2(y)\log\frac{(1-\lambda)p_2(y)}{(1-\lambda)q_2(y)} \\
&= \lambda p_1(y)\log\frac{p_1(y)}{q_1(y)} + (1-\lambda)p_2(y)\log\frac{p_2(y)}{q_2(y)}.
\end{aligned}$$

Summing over all $y \in \mathcal{Y}$:

$$\begin{aligned}\mathrm{KL}(\lambda p_1 + (1-\lambda)p_2 \| \lambda q_1 + (1-\lambda)q_2) &\leq \sum_y \lambda p_1(y) \log \frac{p_1(y)}{q_1(y)} + \sum_y (1-\lambda) p_2(y) \log \frac{p_2(y)}{q_2(y)} \\ &= \lambda\, \mathrm{KL}(p_1 \| q_1) + (1-\lambda)\, \mathrm{KL}(p_2 \| q_2).\end{aligned}$$

■

3. **(Inclusive/Exclusive KL)**

   The target distribution is $p(x) = \frac{1}{3}\mathcal{N}(-3,1) + \frac{2}{3}\mathcal{N}(3,1)$. We fit $q(x) = \mathcal{N}(\mu, \sigma^2)$.

   - **Inclusive KL** ($\mathrm{KL}(p\|q)$, forward KL): mode-covering. The optimization finds a q that must cover all modes of p (wherever $p(x) > 0$, $q(x)$ must be non-zero), resulting in a broad distribution centered between the modes. Result: $\mu \approx 1.00$, $\sigma \approx 3.00$.
   - **Exclusive KL** ($\mathrm{KL}(q\|p)$, reverse KL): mode-seeking. The optimization finds a q that concentrates on the dominant mode of p (since q can safely ignore modes). Result: $\mu \approx 3.00$, $\sigma \approx 1.02$.

   We solve both optimizations numerically using gradient descent. The resulting figure is shown in Figure 1.

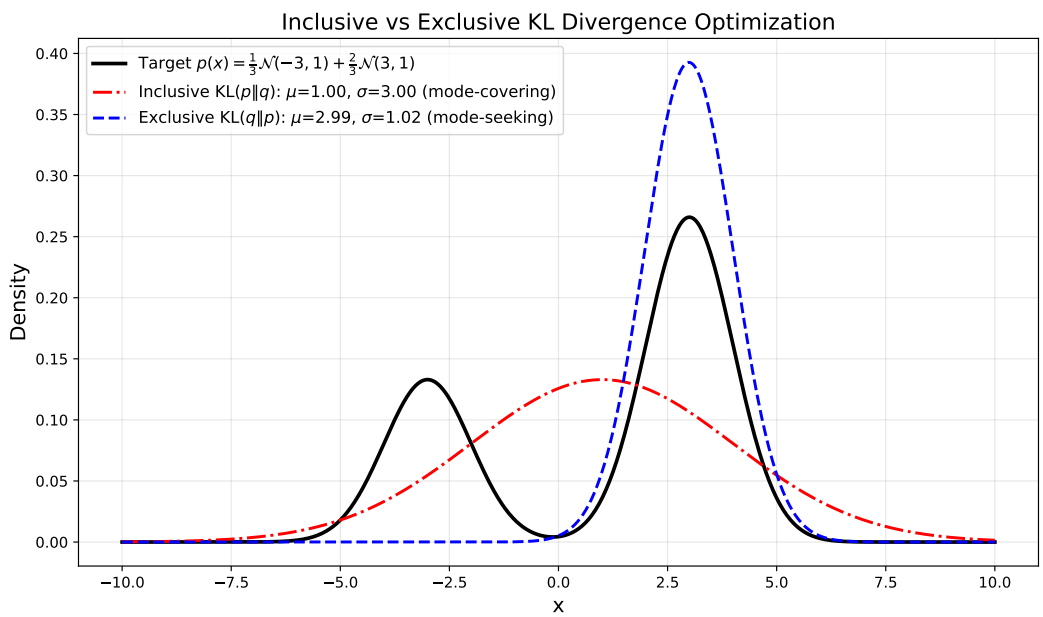

Figure 1: Comparison of inclusive and exclusive KL divergence optimization. The inclusive KL ($\mathrm{KL}(p\|q)$, mode-covering) produces a broad distribution that covers all modes, while the exclusive KL ($\mathrm{KL}(q\|p)$, mode-seeking) concentrates on the dominant mode.

   The Python code used to generate this figure is provided in the accompanying file `kl_divergence.py`.

4. **(Variational Inference with Forward KL)**

   Bob proposes using the forward KL-divergence $\mathrm{KL}(p(z|x)\|q_\psi(z|x))$.

Expanding:

$$\begin{aligned}\mathrm{KL}(p(z|x)\|q_\psi(z|x)) &= \mathbb{E}_{z\sim p(z|x)}\left[\log\frac{p(z|x)}{q_\psi(z|x)}\right]\\ &= \mathbb{E}_{z\sim p(z|x)}[\log p(z|x)] - \mathbb{E}_{z\sim p(z|x)}[\log q_\psi(z|x)].\end{aligned}$$

Since $\mathbb{E}_{z\sim p(z|x)}[\log p(z|x)]$ does not depend on ψ, minimizing over ψ is equivalent to:

$$\max_\psi\ \mathbb{E}_{z\sim p(z|x)}[\log q_\psi(z|x)].$$

This is the objective for $q_\psi(z|x)$: maximize the expected log-likelihood of q_ψ under the true posterior.

Pros:

- The forward KL is *mode-covering* (inclusive): $q_\psi(z|x)$ will try to cover all modes of $p(z|x)$, avoiding mode collapse. This can lead to a more faithful approximation of the full posterior.
- If the true posterior is multi-modal, forward KL will not miss modes.

Cons:

- The objective requires sampling from the true posterior $p(z|x)$, which is intractable in VAE — this is the very distribution we are trying to approximate. Thus, this objective *cannot be directly optimized* in the VAE framework without additional techniques (e.g., importance sampling, MCMC).
- Because forward KL is mode-covering, $q_\psi(z|x)$ may spread probability mass over regions where $p(z|x)$ is negligible, leading to a high-variance, overly diffuse approximation.
- It cannot be incorporated into the standard ELBO framework, which naturally arises from the reverse KL.

Solution 7.

1. **ELBO Derivation**

   We derive the ELBO for $\log p_{\mu,\sigma,\theta}(x)$.

$$\begin{aligned}\log p(x) &= \log\sum_z\int p(x,w,z)\,dw\\ &= \log\sum_z\int\frac{q_{\psi,\phi}(w,z|x)}{q_{\psi,\phi}(w,z|x)}p(x,w,z)\,dw\\ &\geq \mathbb{E}_{(w,z)\sim q_{\psi,\phi}(w,z|x)}\left[\log\frac{p(x,w,z)}{q_{\psi,\phi}(w,z|x)}\right]\quad\text{(by Jensen's inequality)}\end{aligned}$$

   Now expand $p(x,w,z) = p(z)\,p_{\mu,\sigma}(w|z)\,p_\theta(x|w)$ and $q_{\psi,\phi}(w,z|x) = q_\psi(w|x)\,q_\phi(z|w,x)$:

$$\begin{aligned}\mathrm{ELBO} &= \mathbb{E}_{q_\psi(w|x)q_\phi(z|w,x)}\left[\log\frac{p(z)\,p_{\mu,\sigma}(w|z)\,p_\theta(x|w)}{q_\psi(w|x)\,q_\phi(z|w,x)}\right]\\ &= \mathbb{E}_{q_\psi(w|x)q_\phi(z|w,x)}\left[\log p_\theta(x|w) + \log\frac{p_{\mu,\sigma}(w|z)}{q_\psi(w|x)} + \log\frac{p(z)}{q_\phi(z|w,x)}\right]\end{aligned}$$

   Separating the three terms:

$$
\begin{aligned}
\text{ELBO} = & \underbrace{\mathbb{E}_{q_\psi(w|x)}[\log p_\theta(x|w)]}_{\text{Reconstruction}} \\
& + \underbrace{\mathbb{E}_{q_\psi(w|x)q_\phi(z|w,x)}\left[\log \frac{p_{\mu,\sigma}(w|z)}{q_\psi(w|x)}\right]}_{\text{Latent } w \text{ regularization}} \\
& + \underbrace{\mathbb{E}_{q_\psi(w|x)q_\phi(z|w,x)}\left[\log \frac{p(z)}{q_\phi(z|w,x)}\right]}_{\text{Latent } z \text{ regularization}}
\end{aligned}
$$

The three terms can be interpreted as follows:

- **Term 1** (involving $p_\theta(x|w)$): Reconstruction term — measures how well the decoder reconstructs x from latent w.
- **Term 2** (involving $p_{\mu,\sigma}(w|z)$): Can be rewritten as $-\mathbb{E}_{q_\phi(z|w,x)}[\mathrm{KL}(q_\psi(w|x)\|p_{\mu,\sigma}(w|z))]$ — regularizes the encoder $q_\psi(w|x)$ to match the Gaussian mixture component $p_{\mu,\sigma}(w|z)$.
- **Term 3** (involving $p(z)$): Equals $-\mathbb{E}_{q_\psi(w|x)}[\mathrm{KL}(q_\phi(z|w,x)\|p(z))]$ — regularizes the cluster assignment $q_\phi(z|w,x)$ to match the prior $p(z)$.

2. **Training Procedure for GM-VAE**

   (a) **Network Architecture:**
   - **Encoder** $q_\psi(w|x)$: A neural network that takes x as input and outputs the mean $\mu_\psi(x)$ and variance $\sigma^2_\psi(x)$ of a Gaussian distribution over w. Use the reparameterization trick: $w = \mu_\psi(x) + \sigma_\psi(x) \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$.
   - **Cluster inference network** $q_\phi(z|w,x)$: A neural network that takes w and x as input and outputs a categorical distribution over $z \in \{1, \ldots, K\}$ (softmax output).
   - **Decoder** $p_\theta(x|w)$: A neural network that takes w as input and outputs the mean $\mu_\theta(w)$ and variance $\sigma^2_\theta(w)$ of a Gaussian distribution over x.
   - **Learnable prior parameters**: μ_k, σ_k for $k = 1, \ldots, K$ (means and standard deviations of each Gaussian component).

   (b) **Training Loop** (for each mini-batch of data $\{x^{(i)}\}$):
   i. **Encode**: For each $x^{(i)}$, compute $\mu_\psi(x^{(i)})$ and $\sigma_\psi(x^{(i)})$, sample $w^{(i)} = \mu_\psi(x^{(i)}) + \sigma_\psi(x^{(i)}) \odot \epsilon^{(i)}$ via reparameterization.
   ii. **Cluster Assignment**: Compute $q_\phi(z = k|w^{(i)}, x^{(i)})$ for each $k = 1, \ldots, K$.
   iii. **Decode**: Compute $\mu_\theta(w^{(i)})$ and $\sigma^2_\theta(w^{(i)})$ to evaluate $\log p_\theta(x^{(i)}|w^{(i)})$.
   iv. **Compute ELBO**: Evaluate the three ELBO terms:
   - Reconstruction: $\log p_\theta(x^{(i)}|w^{(i)})$.
   - w-regularization: $\sum_k q_\phi(z = k|w^{(i)}, x^{(i)}) \log \frac{p_{\mu_k,\sigma_k}(w^{(i)}|z=k)}{q_\psi(w^{(i)}|x^{(i)})}$.
   - z-regularization: $\sum_k q_\phi(z = k|w^{(i)}, x^{(i)}) \log \frac{p(z=k)}{q_\phi(z=k|w^{(i)},x^{(i)})}$.

   v. **Update**: Maximize the ELBO using gradient ascent (or equivalently, minimize $-$ELBO) with respect to all parameters $\theta, \psi, \phi, \mu, \sigma$ using Adam or SGD.

   (c) **Key Implementation Details:**
   - Use the reparameterization trick for w to enable backpropagation through sampling.
   - For the discrete variable z, use the Gumbel-Softmax trick or marginalize over z analytically (since z is discrete with K values, we can enumerate all K components).
   - The KL term for z can be computed in closed form as the KL between a categorical and a uniform distribution.

Solution 8. Used Claude (AI assistant by Anthropic) in the following capacities:

- **Solution ideation:** Brainstorming approaches and generating initial ideas for problem-solving strategies, particularly for structuring mathematical proofs and derivations.
- **English grammar and writing:** Polishing and refining the English writing throughout the solutions to improve clarity, coherence, and academic tone.
- **LaTeX formatting:** Optimizing typesetting and layout of mathematical expressions, tables, and document structure for better readability and presentation.
- **Result verification:** Cross-checking the correctness of intermediate computations, final answers, and logical consistency of proofs.

All final solutions were reviewed and verified by the author.